A Fixed Aperiodic Greedy Pair-Sum-Avoiding Sequence

Zhiheng Li

Abstract

We exhibit the fixed 21-element seed

$$A = \{1, 2, 3, 5, 7, 13, 22, 27, 28, 32, 36, 40, 47, 48, 52, 63, 71, 77, 81, 89, 97\},$$

whose greedy pair-sum-avoiding extension has a consecutive-gap sequence that is not eventually periodic. The proof constructs an explicit infinite set S satisfying the exact greedy recurrence $t \in S \iff t \notin S + S$ for $t > 97$, using a scale-eight sumset controller embedded in three residue classes modulo 49 and a finite modular shield.

1 Introduction and the main theorem

AI-use disclosure. This proof was developed with assistance from OpenAI's GPT-5.6 Sol. All arguments were checked and verified by the author.
Formal verification. Theorem 1.1 has been formally verified in Lean 4. The formalization is available at https://doi.org/10.5281/zenodo.21902147.

The question appears in Erdős and Graham [1, p. 53] and is catalogued as Erdős Problem 341 in Bloom's online database [2]. Related periodicity criteria are due to Calkin and Finch [3], while Calkin and Erdős [4] proved that a natural class of aperiodic sum-free sets is incomplete. We answer the arbitrary finite-seed question negatively.

Write $\mathbb{N} = \{1, 2, 3, \ldots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. For $X, Y \subseteq \mathbb{N}_0$ and $c \in \mathbb{N}_0$, put

$$X + Y = \{x + y \mid x \in X,\ y \in Y\}, \qquad c + X = \{c + x \mid x \in X\}, \qquad X^{\mathsf{c}} = \mathbb{N}_0 \setminus X.$$

Equal summands are allowed in every sumset.

Let $A = \{a_1 < \cdots < a_k\} \subseteq \mathbb{N}$ be a nonempty finite set. Install all elements of A as the initial terms. For $n \geq k$, define

$$a_{n+1} = \min\{t \in \mathbb{N} \mid t > a_n,\ t \notin A_n + A_n\}, \qquad A_n = \{a_1, \ldots, a_n\}. \tag{1}$$

Let $d_m = a_{m+1} - a_m$ denote the consecutive gaps.

Theorem 1.1. *Let*

$$A = \{1, 2, 3, 5, 7, 13, 22, 27, 28, 32, 36, 40, 47, 48, 52, 63, 71, 77, 81, 89, 97\},$$

For the sequence generated from A by (1),

$$\forall M \geq 1\ \forall p \geq 1\ \exists m \geq M \quad d_{m+p} \neq d_m.$$

Thus its consecutive gap sequence is not eventually periodic.

Lemma 1.2 (Exact recurrence implies the greedy rule). *Let $S \subseteq \mathbb{N}$ be infinite, let $N \in S$, and put $A = S \cap [1, N]_{\mathbb{Z}}$. Suppose that*

$$t \in S \iff t \notin S + S \qquad (t > N). \tag{2}$$

Then the increasing enumeration of S is exactly the greedy extension of A.

Proof. Scan the integers $t > N$ in order. Inductively, immediately before testing t, the selected set is $S \cap [1, t-1]_{\mathbb{Z}}$. Since all elements of S are positive, every representation $t = u + v$ with $u, v \in S$ has $u, v < t$, and hence

$$t \in (S \cap [1, t-1]_{\mathbb{Z}}) + (S \cap [1, t-1]_{\mathbb{Z}}) \iff t \in S + S.$$

Thus (2) makes the rule select exactly the elements of S, completing the induction. $\square$

2 A scale-eight sumset controller

For integers $a \le b$, write $[a, b]_{\mathbb{Z}} = \{a, a+1, \ldots, b\}$. We use the standard identity

$$[a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}} = [a + c, b + d]_{\mathbb{Z}} \qquad (a \le b,\ c \le d). \tag{3}$$

Define

$$L = \bigcup_{k \ge 0} [8^k, 4 \cdot 8^k - 1]_{\mathbb{Z}} \tag{4}$$

$$Y_0 = \{0\} \cup \bigcup_{k \ge 0} [4 \cdot 8^k, 8 \cdot 8^k - 1]_{\mathbb{Z}}, \tag{5}$$

$$Y_1 = [1, 3]_{\mathbb{Z}} \cup \bigcup_{k \ge 0} [16 \cdot 8^k - 1, 32 \cdot 8^k - 1]_{\mathbb{Z}}, \tag{6}$$

$$Y_2 = [0, 1]_{\mathbb{Z}} \cup [7, 15]_{\mathbb{Z}} \cup \bigcup_{k \ge 0} [64 \cdot 8^k - 1, 128 \cdot 8^k - 1]_{\mathbb{Z}}. \tag{7}$$

At each scale, the corresponding blocks of L and $Y_0 \setminus \{0\}$ partition $[8^k, 8^{k+1} - 1]_{\mathbb{Z}}$. Thus

$$Y_0 = L^{\mathsf{c}}. \tag{8}$$

Lemma 2.1 (Controller identities). *The sets in* (5)–(7) *satisfy*

$$Y_1 = (Y_0 + Y_0)^{\mathsf{c}}, \tag{9}$$

$$Y_2 = (Y_1 + Y_1)^{\mathsf{c}}, \tag{10}$$

$$Y_0 = \{0\} \cup \big(1 + (Y_2 + Y_2)^{\mathsf{c}}\big). \tag{11}$$

Proof. A direct interval calculation gives

$$Y_0 + Y_0 = \{0\} \cup \bigcup_{k \ge 0} [4 \cdot 8^k, 16 \cdot 8^k - 2]_{\mathbb{Z}} = Y_1^{\mathsf{c}}, \tag{12}$$

$$Y_1 + Y_1 = [2, 6]_{\mathbb{Z}} \cup \bigcup_{k \ge 0} [16 \cdot 8^k, 64 \cdot 8^k - 2]_{\mathbb{Z}} = Y_2^{\mathsf{c}}, \tag{13}$$

$$Y_2 + Y_2 = [0, 2]_{\mathbb{Z}} \cup [7, 30]_{\mathbb{Z}} \cup \bigcup_{k \ge 0} [64 \cdot 8^k - 1, 256 \cdot 8^k - 2]_{\mathbb{Z}} = L - 1. \tag{14}$$

Equations (14) and (8) yield (11). $\square$

Lemma 2.2. *The membership indicator of Y_0 is not ultimately periodic.*

Proof. Suppose that there are $N \geq 0$ and $h \geq 1$ such that

$$n \in Y_0 \iff n+h \in Y_0 \qquad (n \geq N).$$

Choose $a = 8^k > \max\{N, h\}$ and put $m = 4a - h$. Then $N < a \leq m \leq 4a-1$. Hence $m \in L$, so $m \notin Y_0$, whereas $m + h = 4a \in [4a, 8a-1]_{\mathbb{Z}} \subseteq Y_0$. This contradicts the assumed periodicity. □

3 A finite shield modulo 49

All residues in this section lie in $\mathbb{Z}/49\mathbb{Z}$ and are represented in $\{0, \dots, 48\}$. Put

$$P = \{7, 14, 28\}, \qquad B = \{3, 22, 32, 40, 48\}, \qquad D = \{1, 2, 5, 13, 27, 36, 47\}, \qquad Q = P \cup B. \tag{15}$$

Lemma 3.1 (Finite shield)**.** *In $\mathbb{Z}/49\mathbb{Z}$,*

$$(B+B) \cup (P+B) \cup (D+B) = \mathbb{Z}/49\mathbb{Z} \setminus Q, \tag{16}$$

$$Q \cap \big((B+B) \cup (P+B) \cup (D+B) \cup (D+P)\big) = \varnothing, \tag{17}$$

$$(P+P) \cap Q = P. \tag{18}$$

Moreover, the only ordered pairs in $P \times P$ whose sum lies in Q are

$$7+7 = 14, \qquad 14+14 = 28, \qquad 28+28 = 49+7. \tag{19}$$

Proof. All assertions follow by direct computation in $\mathbb{Z}/49\mathbb{Z}$. □

We now define the infinite set

$$\begin{aligned} S = D &\cup \{49q + b \mid q \in \mathbb{N}_0,\ b \in B\} \\ &\cup \{49q+7 \mid q \in Y_0\} \cup \{49q+14 \mid q \in Y_1\} \cup \{49q+28 \mid q \in Y_2\}. \end{aligned} \tag{20}$$

Hence

$$S \cap [1, 97]_{\mathbb{Z}} = D \cup B \cup (49+B) \cup \{7, 28, 63, 77\} = A. \tag{21}$$

Theorem 3.2. *For every $t > 97$,*

$$t \in S \iff t \notin S+S. \tag{22}$$

Proof. Write $t = 49q + r$, with $0 \leq r < 49$. Since $t > 97$, we have $q \geq 2$.

First let $r \in Q = B \cup P$. The six possible pair-sum types are

$$D+D, \quad D+B, \quad D+P, \quad B+B, \quad B+P, \quad P+P.$$

A $D+D$ sum is at most $94 < t$. Equation (17) excludes the next four types from Q, while (18) and (19) leave only the three diagonal $P+P$ pairs. Hence

$$\begin{aligned} 49q+14 \in S+S &\iff q \in Y_0 + Y_0, \\ 49q+28 \in S+S &\iff q \in Y_1 + Y_1, \\ 49q+7 \in S+S &\iff q-1 \in Y_2 + Y_2. \end{aligned}$$

Indeed, the corresponding representations are

$$\begin{aligned}
(49u+7)+(49v+7) &= 49(u+v)+14,\\
(49u+14)+(49v+14) &= 49(u+v)+28,\\
(49u+28)+(49v+28) &= 49(u+v+1)+7.
\end{aligned}$$

The exclusion of every other source type proves the converse implication in each equivalence. The controller identities (9)–(11) therefore make every P-rail integer represented exactly when it is absent from S. Every B-rail integer belongs to S, and the same exhaustive source analysis shows that none is represented. Thus (22) holds for $r \in Q$.

Now let $r \notin Q$. Such an integer t is not in S: its residue excludes every infinite rail, and $t > 97$ excludes the finite set D. By (16), there exist $b \in B$ and $x \in B \cup D \cup P$ such that $x + b \equiv r \pmod{49}$. Set

$$e_x = x \quad (x \in B \cup D), \qquad e_7 = 7, \qquad e_{14} = 63 = 49 + 14, \qquad e_{28} = 28.$$

Every e_x belongs to S. Write

$$e_x = 49u_x + x, \qquad x + b = 49c + r.$$

Here $u_x, c \in \{0, 1\}$. Since $q \geq 2$, the quotient $q - u_x - c$ is nonnegative, and

$$t = e_x + \big(49(q - u_x - c) + b\big) \in S + S.$$

The second summand is a positive element of a full B-rail. Both summands are positive and sum to t, so each is smaller than t. This completes the proof. $\square$

By (21), Theorem 3.2, and Lemma 1.2, the increasing enumeration of S is exactly the greedy extension of the seed A in Theorem 1.1.

4 Nonperiodicity of the gaps

Lemma 4.1. *Let $T = \{t_1 < t_2 < \cdots\} \subseteq \mathbb{N}$ be infinite. If the consecutive gaps $t_{m+1} - t_m$ are eventually periodic, then the membership indicator of T is ultimately periodic.*

Proof. Suppose that the gaps have period $p \geq 1$ from index M onward. Let

$$h = \sum_{j=0}^{p-1} (t_{M+j+1} - t_{M+j}) > 0.$$

Every block of p consecutive gaps in the periodic tail has sum h, so $t_{m+p} = t_m + h$ for $m \geq M$. Since T is infinite, it is unbounded. Thus, given $n \geq t_M$, there is a unique $m \geq M$ such that $t_m \leq n < t_{m+1}$, and

$$t_{m+p} \leq n + h < t_{m+p+1}, \qquad (n+h) - t_{m+p} = n - t_m.$$

Thus $n = t_m$ if and only if $n + h = t_{m+p}$, which proves

$$n \in T \iff n + h \in T \qquad (n \geq t_M),$$

This is ultimate periodicity of the membership indicator. $\square$

Proof of Theorem 1.1. The residue-7 rail of S satisfies

$$49q + 7 \in S \iff q \in Y_0 \qquad (q \in \mathbb{N}_0). \tag{23}$$

Suppose, for contradiction, that membership in S were ultimately periodic with period $h \geq 1$. Then $49h$ would also be an ultimate period. For all sufficiently large q, equation (23) would give

$$q \in Y_0 \iff 49q + 7 \in S \iff 49(q+h) + 7 \in S \iff q + h \in Y_0,$$

contradicting Lemma 2.2. Thus the membership indicator of S is not ultimately periodic. The contrapositive of Lemma 4.1 now shows that the consecutive gaps in the increasing enumeration of S are not eventually periodic, which is the assertion of the theorem. □

References

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